

Eric Ding

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EDUCATION

Cornell University

Ph.D., Computer Science, Direction: Systems and Networking, GPA: 4.0

Ithaca, NY, United States

Aug. 2024 – June 2029 Expected

University of Michigan

B.S.E., Computer Science, Summa Cum Laude, GPA: 3.96, Dual Degree Program

Ann Arbor, MI, United States

Aug. 2022 – Apr. 2024

Shanghai Jiao Tong University

B.S.E., Electrical and Computer Engineering, Outstanding Graduate, GPA: 3.76, top 10%

Shanghai, China

Sep. 2020 – Aug. 2024

RESEARCH EXPERIENCE

Graduate Student Researcher

Cornell University

Ithaca, New York

Aug. 2024 – Present

- Designing control plane and simulator support for circuit-switched AI datacenter scale-out.
- Implemented straggler-mitigation collective communication algorithm for distributed ML workloads.
- Developed physical layer routing algorithms for silicon photonic fabrics in rack-scale ML networks.
- Built compute and networking resource scheduler for reconfigurable optical networks based on TPU torus topology.

Machine Learning System Research Assistant

SymbioticLab, University of Michigan

Ann Arbor, MI, United States

May 2023 – Apr. 2024

- Developed *Propius*, a Federated Learning (FL) resource management system, based on a microservice architecture.
- Built and deployed a distributed FL evaluation framework in GPU clusters, supporting multi-job parallel training.
- Implemented *Venn*, an advanced scheduling policy, in *Propius* that improves the average FL job completion time by up to 1.88x compared to random allocation.
- Contributed to open-source project *FedScale*, the largest benchmark for FL, in adaptive FL optimizer implementation.

Embedded System Research Assistant

The Fan Lab, University of Michigan

Ann Arbor, MI, United States

May 2023 – Sep. 2023

- Developed *WASP*, a wireless wearable device, capable of monitoring sweat for early disease detection.
- Designed a reliable communication protocol that operates atop I2C and Bluetooth Low Energy protocols, ensuring high-fidelity data communication between two in-device microcontrollers and a terminal microcontroller.

Machine Learning Theory Research Assistant

John Hopcroft Center, Shanghai Jiao Tong University

Shanghai, China

Sep. 2021 – Mar. 2022

- Studied, implemented, and theoretically analyzed Non-Contrastive Self-Supervised Learning (SSL) algorithms.
- Conducted experiments, and compared the performances of Non-Contrastive SSL and traditional Supervised Learning, showing the robustness of SSL methods on imbalanced datasets with long-tail distribution.

WORK EXPERIENCE

Artificial Intelligence Internship

Bosch

Shanghai, China

May 2024 – Aug. 2024

- Designed and implemented a multi-agent system using LangGraph, OpenAI endpoints, and Neo4J graph database for retrieval augmented generation and domain-specific task reasoning.
- Deployed the system on Microsoft Azure for internal use, achieving an estimated 30% reduction in man-hour.

PUBLICATION

- **Eric Ding**, Chuhan Ouyang, Rachee Singh, Photonic Rails in ML Datacenters, Twenty-Fourth ACM Workshop on Hot Topics in Networks (HotNets 2025) (arXiv:2507.08119)
- Arjun Devraj, **Eric Ding**, Abhishek Vijaya Kumar, Robert Kleinberg, Rachee Singh, Efficient AllReduce with Stragglers, Under submission (arXiv:2505.23523)
- Abhishek Vijaya Kumar, **Eric Ding**, Arjun Devraj, Darius Bunandar, Rachee Singh, Morphlux: Programmable chip-to-chip photonic fabrics in multi-accelerator servers for ML, Under submission (arXiv:2508.03674)
- **Eric Ding**, Rachee Singh, PipSwitch: A Circuit Switch Using Programmable Integrated Photonics, Optical Fiber Communication (OFC) Conference 2025 (arXiv:2501.18136)
- Jiachen Liu, Fan Lai, **Ding Ding**, Yiwen Zhang, Mosharaf Chowdhury, Venn: Resource Management Across Federated Learning Jobs, MLSys 2025 (arXiv:2312.08298)
- Anjali Devi Sivakumar, Ruchi Sharma, Chandrakalavathi Thota, **Ding Ding**, Xudong Fan, WASP: Wearable Analytical Skin Probe for Dynamic Monitoring of Transepidermal Water Loss, ACS Sensors 2023.

VOLUNTEERING

- SIGCOMM 2025 AE Committee, *Aug. 2025*
- MLSys 2025 AE Reviewer, *Mar. 2025*
- Lunch and Learn Czar, *Computer Science Department, Cornell University, Feb. 2025 - May. 2025*
- Computer Science Peer Tutor, *School of Information, University of Michigan, Feb. 2024 - May. 2024*

TEACHING EXPERIENCE

- TA, Systems for Large-scale ML (CS5470), Cornell University, Fall 2025

GRANTS AND AWARDS

- Graduate School Fellowship, Cornell University, Feb. 2024
- James B. Angell Scholar, University of Michigan, Feb. 2024
- University Honors, University of Michigan, Dec. 2023, Apr. 2023, Dec. 2022
- Dean's List, University of Michigan, Dec. 2023, Apr. 2023, Dec. 2022
- Tang Junyuan JI Scholarship Nominee, Aug. 2022
- Shanghai Jiao Tong University Merit Student Award, Nov. 2021

TECHNICAL SKILLS

Languages: C++, Python, C, SystemVerilog, Go, RISC-V, Matlab, Javascript, Bash, SQL, R

Frameworks: CUDA, NCCL, PyTorch, Tensorflow, JAX, vLLM, DeepSpeed, TorchTitan, LangChain, React, gRPC, Hadoop

Developer Tools: Git, Docker, Kubernetes, Redis, Nvidia Nsight, L^AT_EX, Neo4J, VSCode, Arduino, STM32CubeIDE, Wireshark, Gurobi

Simulation and Modeling: Verdi, Catia, Matlab, Mathematica, LabVIEW, Pspice, Proteus, Vivado